

Mechatronics Design Project: Wind Turbine/Generator and AC-to-DC Power Conversion

Background Information

A generator is a device that converts mechanical energy into electrical energy for external use. Sources of mechanical energy include steam turbines, gas turbines, water turbines, internal combustion engines, and wind turbines.

For this project, you will be required to develop a wind turbine system. A wind turbine system is one type of mechatronic device whose operation is based on Faraday's law of electromagnetic induction. That is, they operate through the interaction of a magnetic field and an electric current. As this process is reversible, the same device can be used as an **electric motor** that converts electrical power into mechanical power, or as a **generator** that converts mechanical power into electrical power.

Project Overview

- You will be working in 2 or 3 person groups (3 is encouraged). All students are expected to work equally on the project.
- Project group partners should be in the same lab session.
- You are required to come to lab each week during your regular lab time to work on your project and check in with the TA on your project progress. Your TAs are there to act as "consultants" as you work on your project.
- The project deliverable consists of demonstrating your system to the TA and creating a **deck of slides as a report** that describes your system and how it works.



Components provided

- Generator
- Propeller blades (we provide a baseline option. If you're interested, you may 3D print your own propeller in the makerspace, but this is not required)
- Electronic components (resistors, diodes, comparators, transistors, LEDs in various colors, etc.)

Components to be designed and fabricated on your own

- Circuit for control of indicator lights
- Mounting for generator (you may use the makerspace, but it is not required)

Deliverables

- Fully functional windmill and indicator system
- Oral presentation to TA
- PowerPoint slides submission

Required System Functionality

1. Convert the AC voltage output from the generator to a DC voltage
2. Build an indicator system that features the following:
 - an indicator light to show that the windmill is not spinning fast enough
 - an indicator light to show that the windmill is spinning too quickly
 - an indicator light whose intensity increases as the windmill speed increases

Design a procedure to collect experimental data for three different wind speeds. Use what you learned this semester, **especially in Labs 4, 5, and 6**, to guide you in your experimental data collection/formatting, and the design of your indicator circuits.

Your presentation and slides should feature the following:

- circuit design overview, components used, and a description of how the system works
- circuit diagram and photo of setup
- experimental data
- experimental evaluation – did your system work as expected?
- difficulties encountered, lessons learned (what would you do differently? what mistakes were made?), conclusions

Extra credit opportunities

- Build a more “granular” indicator system, i.e., in addition to indicator lights that show “too slow” or “too fast”, also include one or two additional indicator lights that indicate that the windmill speed is within the acceptable range, e.g. if you add one extra light, this would indicate “within acceptable range”; if you add two extra lights, one can indicate “acceptable low speed” and the other “acceptable high speed”.
- Any other **ELECTRICAL** feature that you can think of.
- You will not get extra points for a fancy generator mounting or propeller. However, if you want to do this for your own enjoyment/experience, feel free to do so – make sure to use your time wisely!

Timeline**Week 1**

- Come to your regularly scheduled lab session to: (1) experiment with the generator to get a better understanding of its operation, and (2) consult with the TAs so that you better understand the project and what you need to do.
- Plan your control and indicator system – it’s best to build a model of your system in Multisim first.

Week 2

- Come to your regularly scheduled lab session to work on project and consult with TAs
- **Informal mid-project check-in** with TAs to ensure that adequate progress is being made

Week 3

- Come to your regularly scheduled lab session to work on your project and consult with TAs. **All of your laboratory work should be completed by this time!** You will not have access to the lab before the project is due.
- Make a PowerPoint Presentation about your project. Submit an electronic copy of your PPT on Canvas as a report.

Week 4:

- Come to your regularly scheduled lab session – with your PPT – to present your project and demonstrate the operation of the system to your TA.